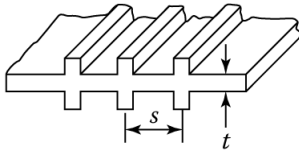


Q1: A simply supported square plate of sides a is subjected to a uniform load p_0 . The plate is reinforced by single equidistant stiffeners (Table 8.1 - Case B). What should be the value of p_0 for an allowable deflection $w_{\max} = 1 \text{ mm}$? Retain only the first term of the series solution. ($E = 200 \text{ GPa}$, $\nu = 0.3$, $s = 200 \text{ mm}$, $t = 20 \text{ mm}$, and $I = 12 \times 10^{-7} \text{ m}^4$)

Solution

First, we refer to Table 8.1 to find the equations for calculating D_x , D_y , and H .

B. Plate reinforced by equidistant stiffeners



$$D_x = H = \frac{Et^3}{12(1-\nu^2)} \quad D_y = \frac{Et^3}{12(1-\nu^2)} + \frac{E'I}{s}$$

E, E' : Elastic modulus of plating and stiffeners, respectively

ν : Poisson's ratio of plating

s : Spacing between centerlines of stiffeners

I : Moment of inertia of the stiffener cross section with respect to midplane of plating

Figure 1: Case B of Table 8.1, Ugural

From this table, we obtain:

$$D_x = H = \frac{Et^3}{12(1-\nu^2)} \quad (1a)$$

$$D_y = \frac{Et^3}{12(1-\nu^2)} + \frac{E'I}{s} \quad (1b)$$

By substituting the material properties from the given data into these two equations, we obtain:

$$D_x = H = 146.52 \text{ kN.m}, \quad D_y = 1346.52 \text{ kN.m}$$

Since all edges of the plate are simply supported, we use Navier's approach to solve the problem. Consider following series for deflection and loading.

$$w(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} w_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (2)$$

$$P(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} P_{mn} \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} \quad (3)$$

Referring to the reference notes, by substituting equations (2) and (3) into the governing differential equation for anisotropic plates, we obtain:

$$D_x \frac{\partial^4 w}{\partial x^4} + 2H \frac{\partial^4 w}{\partial x^2 \partial y^2} + D_y \frac{\partial^4 w}{\partial y^4} = p \quad (4)$$

(5)

As instructed to consider only the first term of the series, we set $m = n = 1$. Since the plate is square, $a = b$. Thus,

Deflection equation simplifies to:

(6)

The maximum deflection occurs at the center of the plate at $x = y = \frac{a}{2}$.

(7)

Finally, substituting values in (7):

$$p_0 = 111.63 \frac{\pi^6}{a^4}$$

Q2: An edge-supported hemispherical container is filled with a liquid of specific weight γ . Demonstrate that the membrane stresses are given by

$$\sigma_\phi = \frac{\gamma a^2}{3t} \left(1 + \frac{\cos^2(\phi)}{1 + \cos(\phi)} \right), \quad \sigma_\theta = \frac{\gamma a^2}{3t} \left(2 \cos(\phi) - \frac{1}{1 + \cos(\phi)} \right)$$

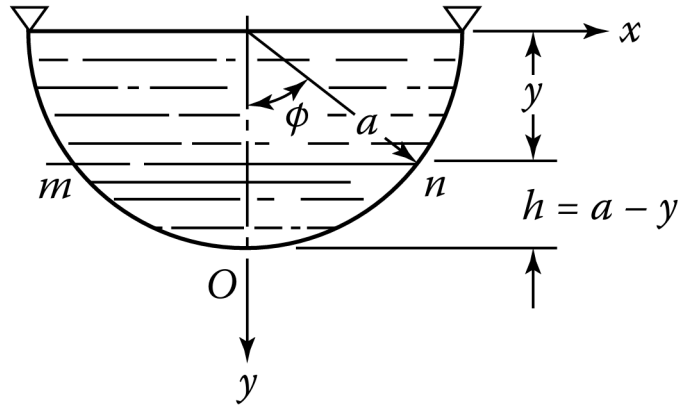


Figure 2: An edge-supported hemispherical container

Solution

First, we need to find the equation for the volume. We consider the total volume to consist of two parts: the cylinder volume and the spherical cap volume.

$$V_c = \pi r_o^2 y; \quad r_o = a \sin \phi, \quad y = a \cos \phi \quad (8)$$

$$V_s = \frac{1}{3} \pi h^2 (3a - h); \quad h = a - y = a(1 - \cos \phi) \quad (9)$$

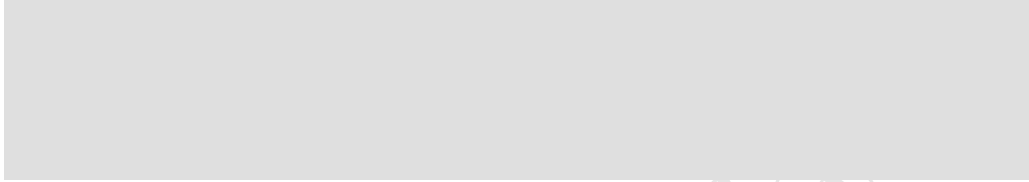
Then,

$$(10)$$

According to the equilibrium equations for a spherical shell,

$$N_\phi = \frac{-F}{2a\pi \sin \phi^2} \quad (11)$$

Now, substituting (10) into (11),

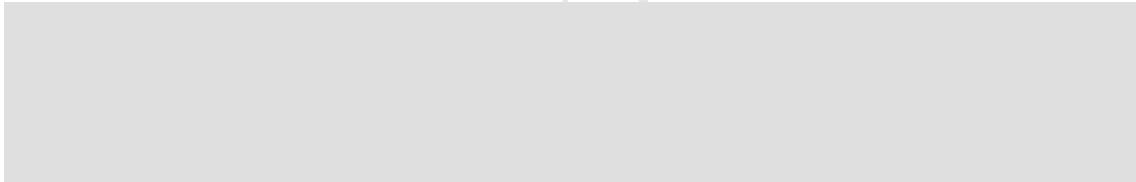


And from $\sigma_\phi = \frac{N_\phi}{t}$, we obtain

$$\sigma_\phi = \frac{\gamma a^2}{3t} \left[1 + \frac{\cos \phi^2}{1 + \cos \phi} \right] \quad (12)$$

To find N_θ , we use the relation:

$$N_\phi + N_\theta = -p_z a; \quad p_z = -\gamma y \quad (13)$$



And from $\sigma_\theta = \frac{N_\theta}{t}$, we obtain

$$\sigma_\theta = \frac{\gamma a^2}{3t} \left[2 \cos \phi - \frac{1}{1 + \cos \phi} \right] \quad (14)$$

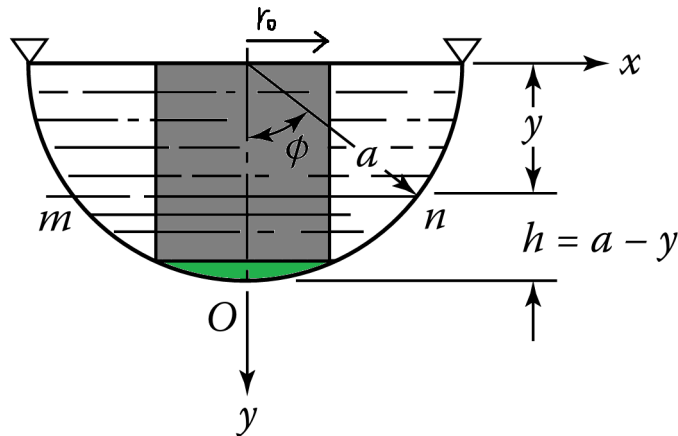


Figure 3: An edge-supported hemispherical containe